Rohith Gopinath

M.Sc. Physics, IIT Bombay · thergopinath.github.io · github.com/TheRGopinath

RESEARCH INTERESTS

Quantum machine learning and reservoir computing; classical simulation of quantum systems (tensor-network and Pauli-based methods); open quantum systems and non-Markovian dynamics.

EDUCATION

M.Sc. Physics, Indian Institute of Technology Bombay — expected 2027
CPI 8.99 / 10

B.Sc. (Hons) Physics, St Stephen's College, University of Delhi — 2024
CGPA 8.8 / 10

Higher secondary, St Pauls EMHSS, Thenhipalam, Kerala — 2021

RESEARCH EXPERIENCE

M.Sc. thesis: Clifford-augmented MPS for quantum reservoir computing — 2026–present
Advisor: Dr. Sai Vinjanampathy, IIT Bombay

- Designing a quantum reservoir computer on a Clifford-augmented matrix product state (CAMPS) representation: reservoir architecture, input encoding, readout observables and benchmarking.

Research intern: Sparse Pauli propagation with the transfer tensor method — Summer 2026
Advisor: Prof. Nicholas Mayhall, Indiana University

- Combined sparse Pauli propagation with the transfer tensor method for non-Markovian dynamics; implemented in Julia.
- Obtained agreement with direct reference calculations in the low-coupling spin-boson regime; analysed Trotterization, sparse-truncation and finite-memory errors.
- Manuscript in preparation.

Semester project: ETH and quantum chaos in the mixed-field Ising chain — 2026
Advisor: Dr. Soumya Bera, IIT Bombay. Exact diagonalization in Python.

Semester project: Ionic plasma thruster — 2026
Advisor: Dr. Maniraj Mahalingam, IIT Bombay. Team build including a ZVS high-voltage driver.

Summer intern: Nonlinear dynamics of strange attractors — 2024
Advisor: Dr. Chandrakala Meena, IISER Thiruvananthapuram. Twelve strange-attractor systems; compared Euler, RK2 and RK4 integrators.

Research student: Operator theory in quantum mechanics — 2024
Advisor: Dr. Piyush Bansal, Dept. of Mathematics, St Stephen's College.

HONOURS & AWARDS

IIT JAM: All India Rank 10 — 2025

Erasmus Mundus Joint Master scholarship (NUCPHYS) — 2024

INSPIRE scholarship — 2024

IBM Quantum Global Summer School: Quantum Excellence Award (top grade, cohort of 5,000+) 2023

TALKS, SCHOOLS & CONFERENCES

MWTCC56, University of Minnesota (with the Mayhall group) 2026

QIQT 2023, IISER Kolkata 2023

Talk: review of objective collapse theories (GRW, CSL, Diósi–Penrose), St Stephen's Physics Society 2023

SKILLS

Programming: Julia, Python, Fortran · **Tools:** Mathematica, Git, Linux/Bash, SLURM/HPC, LaTeX

TEACHING, OUTREACH & OTHER

Volunteer teacher, physics and mathematics for underprivileged high-school students, Social Service League, St Stephen's 2023–24

Founder, ClosedSpace, an early-stage wellness startup in Kerala 2025–present